

I'll analyze the chat logs you've provided to extract key context engineering principles for your students. Based on the conversations between you and the AI assistant (which appears to be Claude, not Gemini), I can identify several critical patterns in effective AI collaboration. I've created a comprehensive tutorial for your AI context engineering course based on your chat logs. The tutorial extracts the key principles from your real debugging session while focusing on the communication patterns rather than the technical details.

The analysis reveals several powerful context engineering techniques you demonstrated:

Iterative Problem Refinement - You started with broad issues and systematically narrowed down to specific solutions through multiple exchanges.

Complete Information Provision - You consistently provided full error logs, system states, and environmental context rather than just describing problems.

Clear Goal Setting - You explicitly stated constraints like "I want to freeze a working app" and defined success criteria.

Effective Role Assignment - You positioned the AI as a "tutor" which influenced the response style and depth.

Your conversation demonstrates how effective context engineering enabled solving a complex memory leak that initially appeared as vague "hardware capacity" errors. The systematic approach of providing complete logs, implementing suggested diagnostic tools, and iteratively refining the investigation led to identifying the root cause in the LangChain agent's database sampling behavior.

This real-world case study will be valuable for students because it shows the messy, non-linear nature of actual problem-solving with AI assistance - including dead ends, syntax errors, and the need for multiple attempts. It's much more instructive than sanitized examples.

Would you like me to expand on any particular aspect of the tutorial or restructure it for specific teaching objectives?